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Conveyor Drive Sizing and Overload Check Worksheet

For maintenance and design engineers working with belt conveyors driven by geared DC motors

Fill in Step 1 from drawings and site measurements, then work through the calculations. Complete the mechanical checks in Step 4 before deciding to fit a larger motor: many overloaded conveyors are cured by maintenance alone.

Step 1 — Record the conveyor data

- [ ] Conveyor length (m) and incline angle (°).
- [ ] Belt speed (m/s) and drive drum diameter including lagging (mm).
- [ ] Total belt mass (kg) and rotating mass of all rollers and idlers (kg).
- [ ] Maximum product mass on the belt at once (kg) = mass per item × items on the belt.
- [ ] Bed construction and friction coefficient: slider bed 0.25–0.45, roller bed 0.03–0.05, troughed idlers 0.02–0.03.
- [ ] Gear ratio, gearbox efficiency, motor continuous torque, rotor inertia and controller current limit.
- [ ] Supply voltage at the panel and at the motor terminals during a loaded start.

Step 2 — Running pull and motor torque

- [ ] Friction pull (N) = friction coefficient × 9.81 × (product + belt + roller mass) × cos(incline).
- [ ] Lift pull (N) = product mass × 9.81 × sin(incline). Belt mass cancels between carrying and return strands.
- [ ] Drum torque (Nm) = total pull × drum diameter (m) / 2.
- [ ] Motor torque (Nm) = drum torque / (gear ratio × gearbox efficiency).
- [ ] Drum speed (rpm) = 60 × belt speed / (3.1416 × drum diameter); motor speed = drum speed × ratio.
- [ ] Power at the drum (W) = total pull × belt speed.
- [ ] Check: running motor torque ≤ 80 % of the continuous rating.

Step 3 — Loaded restart

- [ ] Breakaway friction = 1.3–1.5 × running friction on roller beds; allow more on slider beds.
- [ ] Acceleration force (N) = (product + belt + roller mass) × belt speed / ramp time.
- [ ] Rotor term (Nm) = rotor inertia × motor speed (rad/s) / ramp time.
- [ ] Restart motor torque = (breakaway friction + lift + acceleration force) × drum radius / (ratio × efficiency) + rotor term.
- [ ] Check: restart torque ≤ 80 % of the torque available at the controller current limit.
- [ ] If restart fails but running torque is within rating, lengthen the ramp before upsizing.

Step 4 — Mechanical checks before upsizing

- [ ] Empty-belt current against the commissioning value: a large rise means mechanical drag.
- [ ] Walk the running conveyor for stationary rollers, polished flats and belt-edge wear; scan bearings with an IR thermometer.
- [ ] Check belt tracking, skirt rubber pressure and scraper adjustment.
- [ ] Accumulation zones: add product-to-belt friction (0.3–0.6) × queued mass × 9.81 to the pull.

- [ ] Slip test: chalk-mark belt and drum; if the drum turns while the belt stops, fix tension, not the motor.
- [ ] Slack-side tension ≥ pull / 1.57 for 180° wrap on a lagged drum with belt-to-drum friction 0.3.
- [ ] Terminal voltage drop during a loaded start: investigate cabling if more than about 5 %.

Step 5 — Select the fix and protect the drive

- [ ] Re-ratio only if the lower belt speed still meets the throughput target.
- [ ] Larger geared motor: continuous rating ≥ 1.25 × running motor torque at the required speed.
- [ ] Recalculate the restart with the new motor's rotor inertia.
- [ ] Gearbox output rating ≥ restart drum torque × service factor for the duty.
- [ ] Inclines: run-back torque at the motor = lift pull × drum radius / ratio; fit a fail-safe brake or backstop well above it.
- [ ] Set soft-start and stop ramps and a current limit within the motor's permitted peak current.

Worked example from the article: 6 m roller-bed conveyor at 12°, 0.30 m/s, 150 mm drum, 60:1

Case	Belt pull	Motor torque	Against limit	Result
15 kg sacks, running	235 N	0.35 Nm	77 % of 0.45 Nm rating	Worked at commissioning
25 kg sacks, running	380 N	0.56 Nm	124 % of rating	Overheats
25 kg, restart in 0.5 s	533 N	0.88 Nm	110 % of 0.80 Nm limit	Stalls
25 kg, restart with 2 s ramp	446 N	0.68 Nm	85 % of limit	Starts, still overheats

Friction coefficients are typical design ranges; use belt and roller manufacturers' data where available. Confirm motor, gearbox and brake ratings against the supplier's datasheets. Globe Scott engineers can review your conveyor data through www.globescott.net/contact-us/.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.